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The Stratified Geography of Authorship: A U-Shaped Relationship Between Journal Influence Tier and Author-Country Concentration in Sociology

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ABSTRACT

This study contributes to the literature on global inequality in scientific knowledge production by examining how author-country concentration varies with journals' positions in the academic publishing hierarchy. Using a longitudinal dataset of sociology journals from the Web of Science and Journal Citation Reports covering 1997–2022, we distinguish between differences across journals and changes within the same journal over time. The between-journal analysis reveals a U-shaped association between journals' average Journal Impact Factor and average author-country concentration: concentration is higher among journals at the lower and upper ends of the influence hierarchy than among journals occupying intermediate positions. Two-way fixed-effects models provide complementary evidence of a similar nonlinear association between annual changes in Journal Impact Factor and changes in author-country concentration within journals. Together, these findings indicate that the association between journal influence and the geographic concentration of authorship is not adequately summarized by a single monotonic trend and provide a more differentiated empirical account of inequality in global scientific knowledge production.

1 | Introduction

Scientific knowledge is widely regarded as a cumulative and collective endeavour that benefits from intellectual, cultural and geographic diversity. Scholars from different national backgrounds bring heterogeneous perspectives, research agendas and methodological traditions, thereby enriching scientific debate and fostering innovation (Nielsen et al. 2017; Hofstra et al. 2020). From this perspective, the globalization of science should, in principle, be accompanied by an increasingly diverse geography of knowledge production. Yet, empirical evidence consistently reveals a starkly uneven global distribution of scientific authorship. A disproportionate share of publications in influential journals continues to originate from a small number of countries, predominantly located in the Global North (Smith

et al. 2023; Zumel Dumlaio and Teplitskiy 2025). Rather than moving towards a more balanced global landscape, scientific publishing appears to reproduce, and in some cases intensify, long-standing hierarchies in the production and recognition of knowledge. Although scholars consistently argue that greater geographic diversity enhances academic progress and facilitates the internationalization of academic fields (Li et al. 2023; Aigner et al. 2025; Mason et al. 2021), the patterns through which diversity and concentration are reproduced within the global publishing system remain insufficiently understood. This limitation constrains our ability to fully grasp the structure and dynamics of global knowledge production.

The persistent tension between the ideal of diversity and the reality of concentration points to a fundamental unresolved

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Key Points

- Author-country concentration varies nonlinearly across the journal influence hierarchy.
- Concentration is higher at the lower and upper ends of the JIF distribution than among intermediate-tier journals.
- Within-journal analyses provide complementary evidence of a nonlinear association between annual JIF and concentration.
- The main pattern is robust across alternative measures and model specifications.

question: how does author-country concentration vary systematically across the stratified hierarchy of academic journals? First, the existing literature offers contrasting directional accounts of the association between journal influence and the geographic composition of authorship. Some studies emphasize the possibility that journals with greater visibility attract authors from a wider range of countries (Li et al. 2023), whereas others emphasize the concentration of authorship within prestigious journals and institutions (Grossmann and Lee 2022). These accounts need not be mutually exclusive if the association varies across different positions in the journal hierarchy. However, limited evidence is available on whether a nonlinear pattern is observed across the full range of journal influence.

Second, much of the existing literature has focused predominantly on top-tier journals (Fontana et al. 2019; Aigner et al. 2025; Aistleitner et al. 2023), while author-country concentration in mid- and lower-tier journals has received less attention. Consequently, it remains unclear whether findings based on elite-journal samples extend across the wider journal hierarchy.

Third, existing studies have provided valuable evidence on journal internationalization and its association with journal characteristics and impact (Li et al. 2023; Moed et al. 2020). However, the literature has not always clearly separated differences across journals from changes occurring within the same journal over time. As a result, it is often unclear whether reported associations reflect stable differences among journals occupying different positions in the hierarchy or changes in author composition accompanying changes within journals. Distinguishing these two levels of variation is important because they represent different empirical relationships and require different interpretations.

This study advances a different conceptual approach. First, rather than treating journal impact as a direct driver of author behaviour, we use Journal Impact Factor (JIF) as an indicator of journals' positions within a stratified global publishing ecology. Average concentration is higher among journals at the lower and upper ends of the JIF distribution than among journals occupying intermediate positions. This finding indicates that a single linear specification does not adequately characterize the observed association across the journal

hierarchy and extends prior research by explicitly evaluating a nonlinear alternative.

Second, using a final panel of 2757 journal-year observations from 150 sociology journals covering 1997–2022, this study broadens the empirical scope of previous work, much of which has focused primarily on elite journals. This broader coverage allows us to compare author-country concentration across the lower, middle and upper portions of the JIF distribution.

Third, we distinguish between two complementary empirical associations. The between-journal model evaluates whether journals occupying different typical positions in the publishing hierarchy differ in their average author-country concentration, whereas the two-way fixed-effects model evaluates whether annual changes in JIF within the same journal are associated with changes in author-country concentration. The two models therefore characterize different aspects of the observed pattern and are interpreted separately.

The paper is structured as follows. Section 1 presents the introduction, followed by the literature review in Section 2. Section 3 describes the data sources, and Section 4 details the research methodology. The results are presented in Section 5, and the paper concludes with a discussion and conclusions in Section 6.

2 | Literature Review

A strong consensus holds that geographic diversity in authorship enhances scientific creativity, rigour and the diffusion of knowledge (Nielsen et al. 2017; Hofstra et al. 2020). Yet, a clear paradox exists at the heart of global academic publishing: leading journals, which benefit from this diversity in their readership and influence, often exhibit strikingly high concentrations of authors from only a handful of developed countries and elite institutions (Smith et al. 2023; Zumel Dumlaio and Teplitskiy 2025; Aigner et al. 2025). This gap between the ideal of diversity and the reality of concentration suggests that the relationship between a journal's influence and the geographic diversity of its authors is more complex than a simple linear association.

While this tension between diversity and concentration is a systemic feature of modern science, this study focuses on the field of sociology. Sociology is a discipline deeply engaged with themes of globalization, inequality and social structures, making the geographic composition of its knowledge producers a subject of intrinsic interest. Furthermore, its well-defined journal hierarchy, ranging from community-oriented publications to journals of global reach, provides an ideal context for examining how varying levels of journal influence are systematically associated with differences in author-country concentration (Collyer 2018; Rodriguez Medina 2019; Shin et al. 2023).

The proposed U-shaped pattern concerns differences across journals occupying different positions in the influence hierarchy. Greater author-country concentration at the lower end of the hierarchy may be associated with journals' regional orientation

and structural barriers affecting their author communities. Journals occupying intermediate positions may be associated with authorship from a broader range of countries, whereas renewed concentration at the top may be consistent with journals' stronger embeddedness in elite academic networks. The following sections develop this logic to formulate a hypothesis of a U-shaped association.

2.1 | Regional Concentration at Lower Tiers and Diversity in the Middle

Low-impact journals often originate from and serve specific geographic or thematic academic communities and may therefore have more regionally concentrated author bases (Boardman 2024; Veretennik and Yudkevich 2023). Their comparatively limited international recognition may be associated with a narrower geographic reach, while structural inequalities in global academic publishing may also shape the range of outlets available to scholars. Scholars in emerging countries may face constraints in access to resources and networks relevant to publishing in high-prestige journals (Burns and Fox 2017; Collyer 2018), and their research topics may be less closely aligned with the agendas commonly emphasized in dominant academic centres (Chen et al. 2022; Hoppe et al. 2019). Prior research also raises the possibility that status-related differences in editorial and peer-review evaluation are associated with unequal access to prestigious journals (Zhang et al. 2026). Taken together, these conditions may be associated with greater reliance on accessible or regionally oriented outlets and may offer a possible explanation for the more concentrated author-country profiles expected among lower-impact journals.

Compared with lower-tier journals, journals occupying intermediate positions may combine sufficient visibility and topical breadth with comparatively broad accessibility (Huang et al. 2025). Prior research suggests that this combination may make them attractive to authors from a wider range of countries and may therefore be associated with lower author-country concentration (Li et al. 2023; Aigner et al. 2025). Accordingly, we expect author-country concentration to be lower among intermediate-tier journals than among journals at the lower end of the hierarchy.

2.2 | Concentration Through Entrenched Academic Power

Journals occupying the top tiers of the influence hierarchy may be more deeply embedded in the global academic power structure. High-impact journals occupy a prominent position in defining scientific quality, while key resources, including editorial board positions, peer-review networks and the capacity to shape intellectual agendas, are often concentrated among elite institutions in a small number of core countries (Zhang et al. 2022; Burgess and Shaw 2010; Carrell et al. 2024).

The literature on homophily suggests that the concentration of editorial roles and peer-review networks within core countries may be associated with closer alignment between these

journals and scholars who share similar academic backgrounds and institutional affiliations (Gaston and Smart 2018). Such network structures may make participation more difficult for scholars located outside established core-country networks (Smith et al. 2023; Zumel Dumlaio and Teplitskiy 2025; Aigner et al. 2025).

The intellectual agendas of these journals may be more closely aligned with theoretical frameworks and research interests dominant in core countries. Manuscripts addressing non-mainstream issues or employing alternative methodologies may be perceived as less rigorous or relevant, not necessarily due to overt bias, but because they fall outside the established paradigm upheld by this power structure (Hou et al. 2011; Reidpath and Allotey 2019). These conditions may place scholars from non-core countries, who may lack the resources, networks or familiarity with dominant research paradigms, at a disadvantage in competing for space in these journals (Melhem et al. 2022).

Paradigmatic alignment, author self-selection and the concentration of editorial resources offer a theoretically plausible explanation for why journals at the upper end of the influence hierarchy may exhibit greater author-country concentration (McKay 2024; Zhang et al. 2025; Carrell et al. 2024). These characteristics may make such journals more closely aligned with scholars embedded in core-country institutional and intellectual networks, while scholars outside those networks may face or perceive greater barriers to participation.

Taken together, these arguments imply a cross-journal pattern. Lower-tier journals may exhibit high author-country concentration in association with their regional orientation and structural barriers affecting their author communities. Intermediate-tier journals may combine sufficient visibility with comparatively broad accessibility and may therefore be associated with authorship from a wider range of countries and lower concentration. Top-tier journals, by contrast, may exhibit renewed concentration consistent with their embeddedness in elite editorial, institutional and intellectual networks. These contrasts lead us to expect a U-shaped association between journals' positions in the influence hierarchy and their author-country concentration.

Hypothesis 1. *Across journals, average author-country concentration is U-shaped with respect to journals' typical positions in the influence hierarchy: concentration is higher among lower- and top-tier journals than among intermediate-tier journals.*

3 | Data

We selected journals in the field of sociology from the Web of Science (WoS) for our empirical analysis for the following reasons. First, sociology is fundamentally engaged in developing general social theories while being deeply rooted in specific national and cultural contexts (Rodriguez Medina 2019; Shin et al. 2023). This inherent tension requires journals to balance publishing locally relevant studies and producing theories considered universally significant. Second, sociology is central to ongoing debates concerning the dominance of Western theoretical frameworks and the marginalization of perspectives from the Global South (Collyer 2018). This makes it a pertinent field

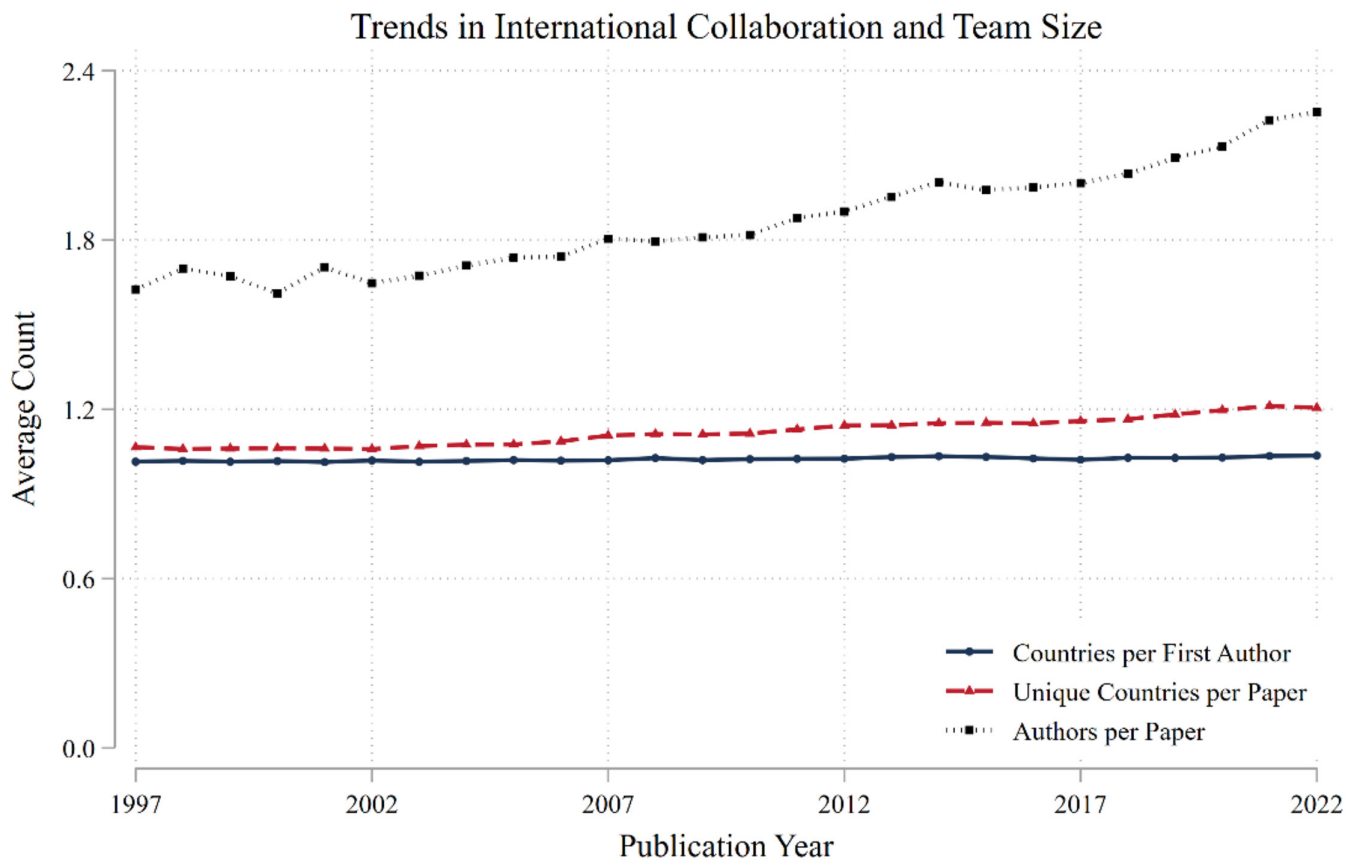


FIGURE 1 | Trends in first authors' international collaboration and team size.

for studying global scientific inequality and the potential concentration of academic influence. Finally, the disciplinary convention of treating the first author as the primary contributor in sociology enables a straightforward and reliable measurement of author-country concentration, a task that would be far more complex in the natural sciences due to their prevalent multi-author traditions (Goyanes et al. 2025).

We collected data from the WoS in November 2025, limiting the retrieval to records categorized under 'Sociology' and classified as articles or reviews. We restricted the analysis to publications from 1997 through 2022 because electronic Journal Citation Reports (JCR) data first became available in 1997 and the inclusion of the Emerging Sources Citation Index in the JCR beginning in 2023 introduced a structural change. After records without a DOI were excluded, the dataset contained 102,791 records. We then examined the 23 pairs of records sharing the same DOI. Twenty pairs represented duplicate WoS records, whereas three involved distinct articles to which WoS had assigned an incorrect DOI. After correcting the three erroneous DOI assignments and removing the 20 duplicate records, the bibliographic corpus contained 102,771 unique articles and reviews. The retained metadata included journal titles, DOIs, publication years, open-access designations, author information and author addresses.

These article-level records were subsequently used to construct the journal-year measures. During this process, 3248 records for which valid country information could not be identified were excluded from the calculation of author-country

concentration. We then standardized journal identities by normalizing capitalization, mapping volume-specific source titles to their parent journals using ISSNs, consolidating historical title variants and excluding four non-journal book series. The resulting cleaned bibliographic dataset covered 152 distinct journals.

We subsequently obtained annual JIFs for journals in the JCR Sociology category and matched them to the publication records by standardized journal title and publication year. JIF is used in this study as an indicator of a journal's relative position within the disciplinary hierarchy. Two journals could not be matched to annual JIF records. After restricting the analysis to journal-years with complete model variables, the final analytical panel contained 2757 journal-year observations from 150 journals over the period 1997–2022. Because the successive data-processing procedures operate at both the article and journal-year levels, the final number of journal-year observations cannot be derived simply by subtracting records with missing country information from the initial article-level corpus.

Figure 1 traces collaboration patterns in sociology from 1997 to 2022 and provides empirical support for our measurement approach. Three trends are evident. First, the dotted line with square markers shows that average team size increased from approximately 1.62 to 2.25 authors per paper. Second, the dashed line with triangle markers shows that the average number of unique countries represented per paper increased from approximately 1.07 to 1.21. Although international collaboration

TABLE 1 | Variable descriptions.

Variable	Definition
Author-country concentration	Herfindahl–Hirschman Index, measuring the concentration of first-author affiliation countries for a journal in a given year.
JIF	Journal Impact Factor, measuring the academic standing of a journal.
Squared centred JIF	The squared term of JIF, introduced to test for a potential U-shaped relationship.
Gold OA ratio	The proportion of a journal's articles published under the gold open access model in a given year.
Total Papers	The annual total number of articles and reviews published by a journal.

increased over the period, these values indicate that most papers continued to involve authors from a single country. Most importantly for our measurement strategy, the solid line with circle markers shows that the average number of countries associated with the first author remained close to one, increasing only from approximately 1.01 to 1.04. This pattern supports the use of the first-author affiliation country in constructing the author-country concentration measure. The complete annual values underlying the figure are reported in Table A1.

4 | Method

4.1 | Dependent Variable

4.1.1 | Author-Country Concentration

We use the Herfindahl–Hirschman Index (HHI) to summarize the distribution of first-author affiliation countries within each journal-year (Herfindahl 1950; Zhang et al. 2021). Following common practice in sociological authorship research, we use the first-author affiliation country to construct the measure (Goyanes et al. 2025). In our sample, 97.55% of papers have first authors associated with a single affiliation country, supporting this measurement choice. For first authors associated with multiple affiliation countries, we retained the country of the first-listed affiliation. Records for which no valid first-author affiliation country could be identified were excluded ($n = 3248$). For each journal-year, we calculated the share of papers associated with each first-author affiliation country and summed the squared shares to obtain the annual HHI:

$$HHI_{jt} = \sum_{i=1}^{n_{jt}} p_{ijt}^2 \quad (1)$$

where p_{ijt} is the share of papers in journal j and year t assigned to first-author affiliation country i , and n_{jt} is the number of distinct affiliation countries represented in that journal-year. Higher HHI values indicate greater author-country concentration, whereas lower values indicate greater dispersion.

Because the conventional HHI estimator may be upwardly biased when country shares are estimated from a small number of articles, we conducted several robustness analyses. We first excluded journal-year observations containing fewer than 5, 10 and 15 articles. We then estimated article-count-weighted fixed-effects models. In addition, we pooled article-level records into

non-overlapping 3-year periods and recalculated the country shares and HHI from the pooled counts. The final period covered 2021–2022 because the observation window ended in 2022. Finally, for journal-years with $N > 1$, we calculated the finite-sample-corrected concentration measure, $(N \times HHI - 1)/(N - 1)$.

4.2 | Independent Variable

4.2.1 | Journal Impact Factor (JIF)

The JIF, which is published in the Journal Citation Reports, is widely used as an indicator of a journal's relative position within the disciplinary hierarchy. The JIF is operationalized as the mean number of citations received in a specific year (e.g., 2023) by articles published in the two previous years (e.g., 2021 and 2022) (Garfield 1999). This calculation, therefore, provides an annual gauge of a journal's standing and visibility within the field. The annual JIF data for all sociology journals were retrieved from the JCR. Furthermore, squared centred JIF is included to test for a nonlinear association with author-country concentration.

4.3 | Control Variables

We control for two key journal-level variables: *Total Papers* and the *Gold OA ratio*. *Total Papers*, which measures a journal's annual output of articles and reviews, is included to account for scale effects. A larger publication volume may be associated with a more geographically diverse author base and, consequently, lower author-country concentration. The *Gold OA ratio* measures the proportion of a journal's articles published as gold open access in a given year (Jia et al. 2026). This variable is controlled because open access availability may simultaneously influence a journal's author distribution and its academic standing. Other commonly used bibliometric indicators, such as the Immediacy Index, Cited Half-Life, Eigenfactor Score and Normalized Eigenfactor, are not included in the model. These metrics are excluded because they are calculated based on data similar to that underlying JIF, namely citation counts and publication volume, which could introduce multicollinearity if included alongside JIF.

Definitions of all variables are provided in Table 1, and descriptive statistics and correlations for all variables are presented in Table 2.

Multicollinearity diagnostics show a maximum variance inflation factor (VIF) of 2.99 and a mean VIF of 2.50, well below common thresholds. These values do not indicate serious multicollinearity.

The distribution of author-country concentration across JIF quartiles reveals a distinct U-shaped pattern, as shown in

TABLE 2 | Correlation matrix and descriptive statistics of the regression variables.

		1	2	3	4	5
1	ln(HHI)	1.00				
2	JIF	−0.12	1.00			
3	JIF squared (after centring)	−0.02	0.72	1.00		
4	Gold OA ratio	0.09	−0.07	−0.00	1.00	
5	ln(Total articles)	−0.26	0.31	0.10	−0.09	1.00
	Mean	−1.003	1.252	1.391	0.046	3.283
	S.D.	0.704	1.175	6.403	0.200	0.722
	Min	−2.950	0.000	0.000	0.000	0.000
	Max	0.000	13.200	145.139	1.000	5.808

Note: $N = 2757$ journal-year observations. Annual JIF is reported in its original scale for ease of interpretation. In the regression models, JIF is centred at 1.153, its mean in the full merged journal-year panel before restriction to the complete-case sample, and the quadratic term is calculated as $(\text{JIF} - 1.153)^2$. The mean JIF in the final estimation sample is 1.252. Author-country concentration and annual article output are reported in logarithmic form, consistent with the regression specifications.

Figure 2. The median HHI does not increase monotonically with JIF rank. Instead, it declines from 0.480 in Q4 (the lowest JIF quartile) to 0.333 in Q2, which represents the trough of the U-shape, and then rises again to 0.446 in Q1 (the highest JIF quartile). This indicates that both the least and most highly positioned journals exhibit higher author-country concentration, while mid-tier journals are the most geographically diverse.

Figure 3 shows the average HHI trends across JIF quartiles over time. The solid red line, representing Q1 journals, fluctuated considerably between 1997 and 2008, with a peak around 0.67–0.70 in 1998–1999. After 2010, however, Q1 HHI began a sustained decline, falling from approximately 0.46 in 2010 to 0.38 in 2022. Q2 and Q3 journals also followed generally decreasing trajectories over the observed period. In contrast, Q4 journals consistently exhibited the highest or second-highest HHI in most years, indicating relatively concentrated authorship in lower-tier journals. The overall downward trend across all quartiles, particularly after 2010, reflects a gradual increase in author diversity within sociology journals. The complete annual values underlying Figure 3, together with the number of journals contributing to each quartile-year mean, are reported in Table A3.

The theoretical argument primarily concerns differences across journals occupying different positions in the publishing hierarchy, whereas a journal fixed-effects model estimates associations using changes within the same journal over time. We therefore estimate two complementary specifications. First, we estimate a between-journal model in which the unit of analysis is the journal. This model relates each journal's mean annual $\ln(\text{HHI})$ to its mean JIF and squared mean JIF over the observation period, while controlling for its mean Gold OA ratio and mean logged annual article count. This specification directly evaluates the cross-journal

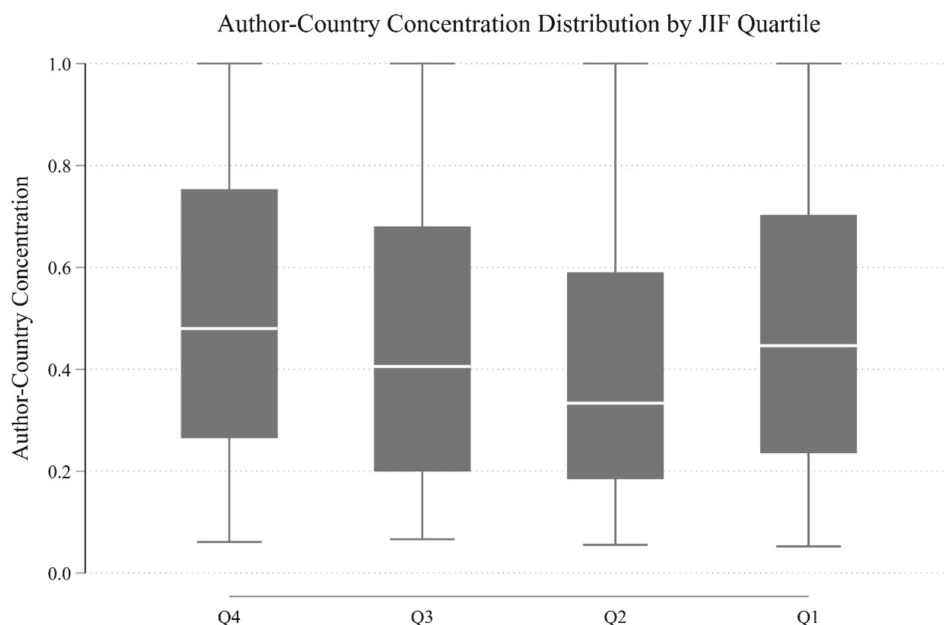


FIGURE 2 | Distribution of author-country concentration by JIF quartile. Q1 denotes the highest JIF quartile and Q4 the lowest. Outlier markers are suppressed for visual clarity. The corresponding distributional statistics are reported in Table A2. The descriptive figures use the 2758 journal-year observations with non-missing HHI and JIF; the regression sample contains 2757 observations after requiring complete control variables.

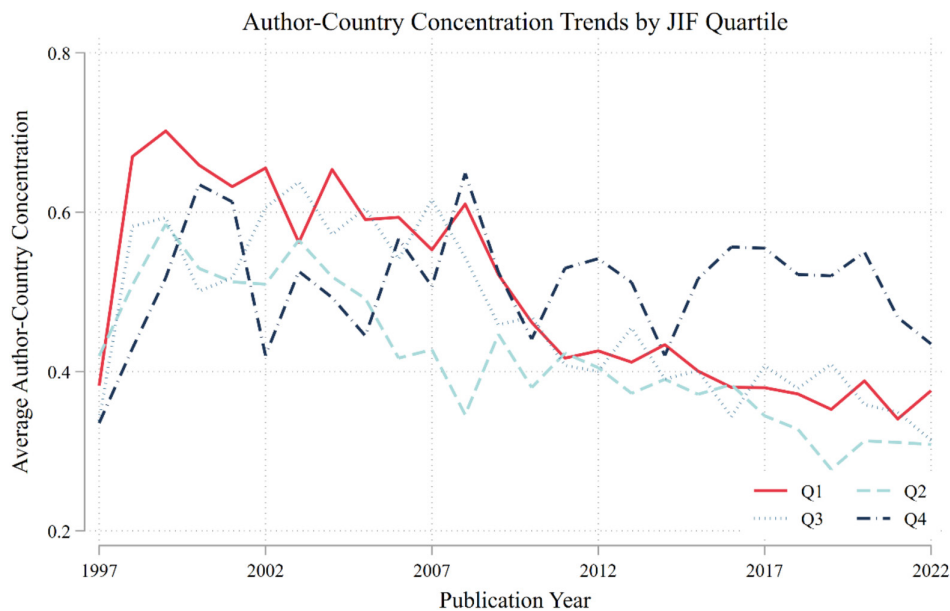


FIGURE 3 | Trends in author-country concentration by JIF quartile over time. Q1 denotes the highest JIF quartile and Q4 the lowest. The annual mean HHI values underlying the figure and the number of contributing journals are reported in Table A3.

hypothesis concerning journals' typical positions in the influence hierarchy.

As an additional between-journal specification, we account for journal specialization using a binary indicator based on JCR subject-category assignments. The indicator equals 1 when a journal is classified exclusively under Sociology and 0 when it is classified under Sociology and at least one additional subject category. It therefore distinguishes journals situated more exclusively within sociology from journals spanning multiple disciplinary categories. Because the indicator is coded as a time-invariant journal characteristic, it is perfectly collinear with the journal fixed effects and cannot be estimated separately in the within-journal models.

Second, we estimate a two-way fixed-effects model to examine whether annual changes in JIF within the same journal are associated with changes in author-country concentration. Journal fixed effects (μ_i) absorb time-invariant differences across journals, including stable journal-specific characteristics, while year fixed effects (λ_t) account for common temporal shocks affecting all journals, such as broader changes in the globalization of sociological research. The resulting estimates therefore represent within-journal associations, net of time-invariant journal characteristics and common year shocks. The between-journal and fixed-effects specifications address distinct empirical quantities and are consequently interpreted separately; neither is treated as direct evidence of the specific mechanisms proposed in the theoretical discussion.

Neither specification resolves the potential endogeneity between JIF and author-country concentration. Although journal fixed effects account for time-invariant journal characteristics and year fixed effects absorb common temporal shocks, they do not eliminate reverse causality or confounding from unobserved time-varying factors. In particular, author-country composition

may itself influence the geographic reach and volume of citations and thereby affect subsequent JIF values. JIF and author-country concentration may therefore be jointly determined. Accordingly, all estimated coefficients are interpreted as conditional associations, and the direction of the relationship cannot be established from our research design.

Because HHI is strictly positive in the analytical sample, we use $\ln(\text{HHI})$ as the main dependent variable. The logarithmic transformation facilitates interpretation of the estimated coefficients in terms of approximate proportional differences. As a robustness check, we estimate the conditional mean of level-form HHI using Poisson pseudo-maximum likelihood (PPML) with journal and year fixed effects. PPML accommodates nonnegative dependent variables and does not require HHI to follow a Poisson distribution. For the within-journal models, annual JIF is centred at 1.153 before constructing its squared term, which reduces the nonessential correlation between the linear and quadratic terms. We also use the logarithm of annual article count to reduce the influence of extreme values. Standard errors in the within-journal models are clustered by journal, whereas the between-journal models use heteroskedasticity-robust standard errors. We estimate two complementary models corresponding to the two levels of variation distinguished above.

$$\overline{\ln(\text{HHI})}_i = \alpha_0 + \alpha_1 \overline{\text{JIF}}_i^c + \alpha_2 (\overline{\text{JIF}}_i^c)^2 + \delta' \mathbf{Z}_i + u_i \quad (2)$$

where $\overline{\ln(\text{HHI})}_i$ is journal i 's mean annual $\ln(\text{HHI})$ over the observation period, and $\overline{\text{JIF}}_i^c$ is its mean annual JIF centred at the sample mean of the journal-level mean JIF values. The vector $\mathbf{Z}_i$ includes the journal-level mean Gold OA ratio and mean $\ln(\text{Total articles})$. In the extended specification, it also includes the time-invariant indicator for whether the journal is classified exclusively under Sociology. The error term is denoted by u_i .

$$\ln(\text{HHI}_{it}) = \beta_0 + \beta_1 \text{JIF}_{it}^c + \beta_2 (\text{JIF}_{it}^c)^2 + \gamma' \mathbf{X}_{it} + \mu_i + \lambda_t + \epsilon_{it} \tag{3}$$

where JIF_{it}^c denotes annual JIF centred at 1.153, and $\mathbf{X}_{it}$ includes the Gold OA ratio and $\ln(\text{Total articles})$ in the controlled specification. Journal fixed effects are represented by μ_i , year fixed effects by λ_t and the idiosyncratic error term by ϵ_{it} .

To assess whether the results depend on the choice of concentration measure, we constructed two alternatives to the HHI. First, we calculated an entropy-based concentration measure as $100 \times [1 - H / \ln(M)]$, where H is Shannon entropy and M is the fixed number of countries represented in the dataset. Second, we calculated the Gini coefficient over the same fixed country universe, assigning zero shares to countries not represented in a given journal-year. Both measures range from 0 to 100, with higher values indicating greater author-country concentration. We re-estimated both the between-journal and journal fixed-effects specifications using these alternative dependent variables. We also replaced the standard JIF with the 5-year JIF in both specifications. Because the 5-year JIF is unavailable before 2007 and is missing for some subsequent journal-years, these models were estimated on the common sample for which the 5-year JIF and all other model variables were available.

5 | Results

5.1 | Between-Journal Association

Table 3 reports the between-journal analysis that directly evaluates the cross-journal hypothesis. The squared centred mean-JIF term is positive and statistically significant ($\beta = 0.129$, $p < 0.001$). The Lind–Mehlum test confirms a U-shaped association over the observed range of mean JIF values ($p = 0.003$): the slope is significantly negative at the lower bound (slope = -0.462 , $p = 0.003$) and significantly positive at the upper bound (slope = 0.695 , $p < 0.001$). The estimated turning point is a mean JIF of 1.792 (95% CI: 1.343–2.242), indicating that average author-country concentration declines across journals below this point and rises across journals above it.

To assess the substantive magnitude of this association, we exponentiated differences in predicted mean $\ln(\text{HHI})$ and expressed them as percentage differences on the original HHI scale. Holding the other covariates constant, journals at the 25th percentile of mean JIF (0.686) had a predicted geometric-mean HHI approximately 17.1% higher than journals located at the estimated turning point. Journals at the 90th percentile of mean JIF (2.231) had a predicted geometric-mean HHI approximately 2.5% higher than journals at the turning point. These estimates indicate that the lower-arm contrast is more pronounced, whereas the upper-arm contrast remains modest at the 90th percentile.

Column (2) adds journal specialization, measured by whether a journal is classified exclusively under Sociology in the JCR. The squared centred mean-JIF term remains positive and statistically significant ($\beta = 0.121$, $p < 0.01$), compared with 0.129 in the baseline specification. Thus, the positive curvature of the between-journal association remains robust after accounting for differences between sociology-only and multidisciplinary

TABLE 3 | Between-journal association between JIF and author-country concentration.

	(1)	(2)
	Baseline	Journal specialization
Mean JIF (centred)	−0.135 (0.085)	−0.112 (0.081)
Squared centred mean JIF	0.129*** (0.034)	0.121*** (0.034)
Mean Gold OA ratio	0.091 (0.138)	0.037 (0.136)
Mean $\ln(\text{Total Papers})$	−0.236*** (0.087)	−0.229*** (0.085)
Sociology-only JCR classification		0.235** (0.102)
Constant	−0.341 (0.286)	−0.453 (0.282)
Observations/Journals	150	150
R^2	0.139	0.171

Note: The unit of analysis is the journal. The dependent variable is each journal's mean annual $\ln(\text{HHI})$ over the observation period. Explanatory variables are calculated as journal-level means using the common estimation sample. Mean JIF is centred at the sample mean before constructing its squared term. 'Sociology-only JCR classification' equals 1 for journals classified exclusively under Sociology and 0 for journals classified under Sociology and at least one additional JCR subject category. Heteroskedasticity-robust standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$.

journals. The specialization indicator is itself positive and statistically significant ($\beta = 0.235$, $p < 0.05$). Conditional on the other journal-level characteristics, sociology-only journals have an estimated geometric-mean HHI approximately 26.5% higher than multidisciplinary journals. This estimate represents a conditional association rather than a causal effect.

5.2 | Within-Journal Fixed-Effects Specification

Table 4 reports complementary panel-data analyses of the association between annual JIF and author-country concentration. The fixed-effects specifications estimate this association using variation within the same journal over time, after accounting for time-invariant journal characteristics and common year shocks. Columns (1) and (2) first examine a linear specification using fixed-effects and random-effects estimators, respectively. The JIF coefficient is statistically insignificant in both models, providing no evidence of a monotonic linear association.

Columns (3) and (4) incorporate both JIF and its squared term. The coefficient on squared JIF is positive and statistically significant at the 1% level in both the fixed-effects model (Column 3) and the random-effects model (Column 4), providing evidence of positive curvature. We formally evaluate the U-shaped pattern for the fixed-effects model in Column (3)

TABLE 4 | Within-journal association between annual JIF and author-country concentration.

	(1)	(2)	(3)	(4)	(5)	(6)
	Linear FE	Linear RE	Quadratic FE	Quadratic RE	Quadratic PPML	Quadratic FE + controls
Centred annual JIF	−0.003 (0.016)	−0.005 (0.016)	−0.048** (0.023)	−0.051** (0.023)	−0.063*** (0.024)	−0.030 (0.021)
Squared centred annual JIF			0.007*** (0.002)	0.007*** (0.002)	0.007*** (0.002)	0.005** (0.002)
Gold OA ratio						0.239** (0.106)
ln(Total Papers)						−0.178*** (0.027)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Journal FE	Yes	No	Yes	No	Yes	Yes
Constant	−1.205*** (0.085)	−1.185*** (0.085)	−1.233*** (0.084)	−1.216*** (0.085)	−0.880*** (0.081)	−0.894*** (0.096)
Observations	2757	2757	2757	2757	2755	2757
Within R^2	0.256	—	0.261	—	—	0.302

Note: Columns (1–4) and (6) use ln(HHI) as the dependent variable. Column (5) uses HHI in its original scale and is estimated using Poisson pseudo-maximum likelihood with journal and year fixed effects. Annual JIF is centred at 1.153 before constructing the quadratic term; therefore, 'squared centred annual JIF' is calculated as $(JIF - 1.153)^2$. Descriptive statistics and correlations for the variables are reported in Table 2. Standard errors clustered by journal are reported in parentheses. Two observations were automatically excluded by `ppmlhdf` when absorbing journal fixed effects; the PPML model is therefore estimated on 2755 journal-year observations. Within R^2 is reported for the fixed-effects models. *** $p < 0.01$, ** $p < 0.05$.

using the Lind–Mehlum (Lind and Mehlum 2010) procedure. The test confirms a statistically significant U-shape over the observed JIF range ($t = 2.35$, $p = 0.010$). The slope is significantly negative at the lower bound (slope = -0.064 , $p = 0.010$) and significantly positive at the upper bound (slope = 0.110 , $p < 0.001$). After converting the centred estimate back to the original JIF scale, the estimated turning point is an annual JIF of 4.825 (95% CI: 3.076–6.575), which falls within the observed range. Figure 4 plots this estimated association, while Table A4 reports the corresponding coefficient estimates, boundary slopes, turning point and formal U-shape test.

To assess its substantive magnitude, we converted differences in predicted ln(HHI) back to percentage differences on the original HHI scale. At the point estimates, an increase in annual JIF from the 25th percentile (0.503) to the estimated turning point is associated with an approximately 11.6% decline in predicted HHI within the same journal, conditional on journal and year fixed effects. This contrast is estimated imprecisely (95% CI: -1.7% to 24.9% ; $p = 0.088$). An increase from the turning point to the 99th percentile of annual JIF (5.422) is associated with an approximately 0.2% increase in predicted HHI (95% CI: -1.1% to 1.5% ; $p = 0.726$). Thus, although the formal test supports a U-shaped association over the full observed range, the effect-size estimates indicate that the upward within-journal contrast remains small and statistically imprecise through the 99th percentile. The visible rise at the extreme upper end should therefore be interpreted cautiously.

Column (5) replaces the dependent variable with the level-form HHI and estimates the model using Poisson pseudo-maximum

likelihood. The positive squared JIF coefficient remains statistically significant, providing additional evidence of positive curvature. Column (6) adds the Gold OA ratio and the logarithm of annual article output to the fixed-effects specification. The squared JIF coefficient remains positive and statistically significant at the 5% level after accounting for journal scale and open-access composition.

Taken together, the between-journal results directly support the hypothesized cross-journal U-shaped pattern: average author-country concentration is higher among journals located at the lower and upper ends of the influence hierarchy than among journals occupying intermediate positions. The fixed-effects results provide complementary evidence of a U-shaped association in changes within the same journal over time. However, the effect-size estimates show that the upward within-journal contrast remains small and statistically imprecise through the 99th percentile. The visible rise in the extreme upper tail, where observations are sparse, should therefore be interpreted cautiously. The between-journal and fixed-effects estimates capture distinct levels of variation and should not be interpreted as interchangeable.

Table A5 reports seven robustness specifications addressing the potential upward bias of HHI in journal-years with small article counts. Across all specifications, the coefficient on squared JIF remained positive and statistically significant. In Panel A, the squared coefficient was 0.004 and significant at the 5% level after excluding journal-years with fewer than 5, 10 or 15 articles. It remained 0.004 but became significant

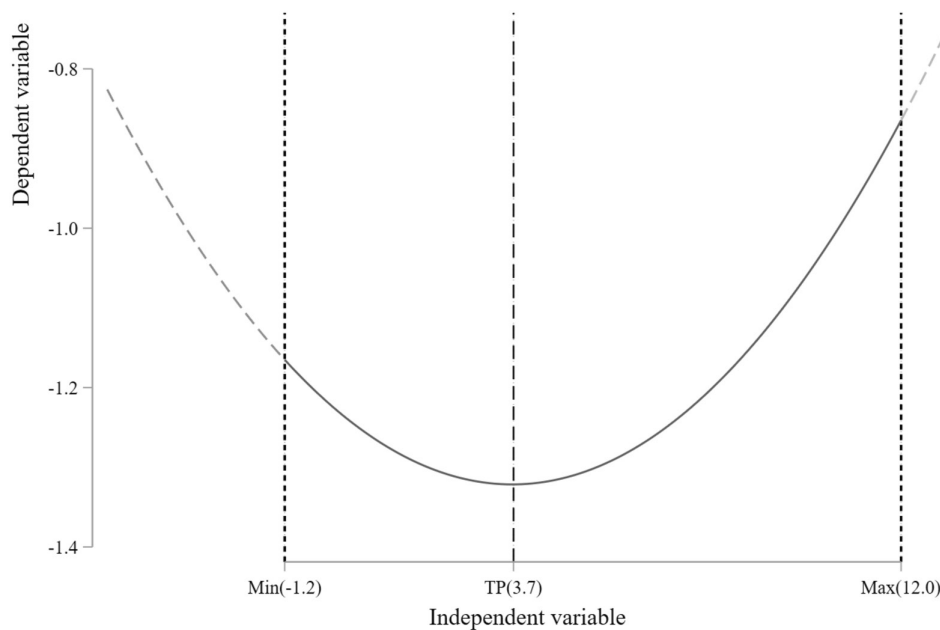


FIGURE 4 | Estimated within-journal association between annual JIF and author-country concentration. The figure is based on the journal fixed-effects specification reported in Column (3) of Table 4. Annual JIF is mean-centred in the estimation and figure. The turning point labelled 3.7 on the centred scale corresponds to an annual JIF of 4.825 on the original scale. The centred minimum and maximum values of -1.2 and 12.0 correspond to original JIF values of approximately 0.0 and 13.2 , respectively. The numerical results underlying the figure are reported in Table A4.

at the 1% level when observations were weighted by article counts. In Panel B, using the finite-sample-corrected HHI produced a positive squared coefficient of 0.002 , significant at the 10% level with the article-count control and at the 5% level without it. When country shares and HHI were recalculated from article-level observations pooled within non-overlapping 3-year periods, the squared coefficient increased to 0.006 and was significant at the 1% level. The linear JIF coefficient was negative in all seven specifications, although it was statistically significant only at the 10% level in the two bias-corrected HHI models. The Gold OA ratio was positively associated with concentration in the annual specifications but was not significant in the pooled 3-year model. Article count was negatively associated with concentration wherever it was included, although its significance weakened to the 10% level in the bias-corrected specification. Overall, the results indicate that the positive curvature of the JIF–concentration relationship is not driven solely by journal-years with small article counts, although the evidence is weaker under the most conservative bias-corrected specification.

We conducted additional analyses to assess the robustness of the results to alternative measures of author-country concentration and journal influence. First, we replaced the HHI with entropy-based concentration and the Gini coefficient (Table A6). The coefficient on squared JIF remained positive and statistically significant in all four models, including both the journal fixed-effects specifications and the between-journal specifications. The positive curvature of the association is therefore not specific to the HHI measure.

Second, we replaced the standard JIF with the 5-year JIF (Table A7). Because the 5-year JIF is unavailable before 2007 and is missing for some subsequent observations, the analysis

was based on a common sample of 1800 journal-years from 147 journals. The squared 5-year JIF coefficient remained positive and statistically significant in both the between-journal model ($\beta = 0.037$, $p < 0.01$) and the journal fixed-effects model ($\beta = 0.003$, $p < 0.05$). These results indicate that the observed positive curvature is not driven solely by the citation window used to construct the standard JIF.

6 | Discussion

6.1 | Summary of Findings

This study addresses the incomplete understanding of the association between journals' hierarchical positions and author-country concentration. Prior studies have emphasized contrasting directional patterns in this association (Grossmann and Lee 2022; Li et al. 2023; Aigner et al. 2025). Rather than assuming that the association is uniformly positive or negative, we examine whether it varies across the JIF distribution. Using a 26-year panel of sociology journals, we find evidence of a statistically supported U-shaped association. The between-journal results indicate that average author-country concentration is higher among journals at the lower and upper ends of the JIF distribution than among journals occupying intermediate positions, while the fixed-effects estimates provide complementary evidence of a nonlinear within-journal association over time.

6.2 | Theoretical and Empirical Contributions

Our primary contribution is empirical: we evaluate whether the association between journal influence and author-country

concentration is adequately represented by a single monotonic relationship. Prior studies have emphasized either geographic dispersion or concentration in different journal samples and contexts. By examining journals across a broader range of influence and explicitly estimating a quadratic specification, our results suggest that negative and positive associations may characterize different portions of the JIF distribution. The identified U-shaped pattern therefore offers a more differentiated empirical account that complements earlier directional findings, without directly identifying the mechanisms underlying the pattern.

Second, this study broadens the available empirical evidence by examining journals across the lower, middle and upper portions of the JIF distribution, extending prior research that has focused primarily on elite outlets. By including lower- and intermediate-impact journals, our analysis documents variation that is not visible in studies limited to elite outlets (Mason et al. 2021). Our results draw attention to the comparatively understudied intermediate portion of the journal hierarchy, where journals exhibit lower average author-country concentration than those at the lower and upper ends of the JIF distribution. By broadening attention beyond elite outlets, this finding provides a more differentiated empirical account of author-country concentration across the sociology journal hierarchy.

Third, our empirical approach distinguishes between two related levels of variation. The between-journal model evaluates differences among journals occupying different typical positions in the influence hierarchy, whereas the fixed-effects model evaluates associations based on changes within the same journal over time. Journal and year fixed effects reduce confounding from stable journal characteristics and common temporal shocks, but they do not render JIF exogenous. Author-country composition may affect citation patterns and subsequent JIF values, while unobserved time-varying journal characteristics may influence both variables. The estimates should therefore be interpreted as associations rather than causal effects, and the direction of the relationship cannot be determined from the present design. Neither specification directly identifies the regional-orientation, differential-accessibility or elite-gatekeeping mechanisms proposed in the theoretical discussion.

6.3 | Implications and Future Research

The U-shaped association documented here provides a more differentiated perspective on inequality in scientific publishing. The relatively low author-country concentration among journals in the intermediate portion of the JIF distribution may reflect their combination of meaningful international visibility and comparatively broad accessibility. These journals also exhibit a more geographically dispersed authorship distribution. At the upper end of the JIF distribution, renewed concentration raises the possibility of an ‘elite ceiling’ associated with prestigious editorial, institutional and peer-review networks, but it does not establish that such a mechanism is operating. Our data do not include submission pools, editorial decisions, reviewer networks or rejection outcomes. Elite gatekeeping should therefore be viewed as a theoretically informed interpretation, not a causally identified mechanism. The observed pattern warrants

closer investigation of whether structural or exclusionary practices contribute to concentration among high-JIF journals.

Future research should build upon this work by directly investigating the mechanisms that may underlie this U-shaped pattern. For instance, qualitative studies could examine whether and how editorial practices and reviewer evaluations in high-impact journals contribute to differential publication outcomes for scholarship from non-core countries. Furthermore, it would be valuable to examine whether this U-shaped relationship holds in other academic disciplines, particularly those in the natural sciences, which may have different collaborative and publishing cultures.

6.4 | Conclusion

In conclusion, our findings indicate that the association between journal position, as proxied by JIF, and author-country concentration is not adequately described as uniformly positive or negative. Within the sociology journals examined here, average concentration is higher at the lower and upper ends of the JIF distribution and lower among journals occupying intermediate positions. This U-shaped pattern provides a more differentiated account of geographic inequality across the sociology journal hierarchy.

6.5 | Limitations

This study has several limitations that should be acknowledged. First, by focusing exclusively on sociology, the generalizability of our findings to other disciplines, such as the natural sciences or other social sciences, remains to be tested. Second, consistent with common practice in sociological research, our measure of author-country concentration is based solely on the first-author affiliation country. While this approach offers clarity, it cannot capture the full complexity of international collaboration among co-authors, nor does it reflect institutional diversity or regional variation within countries. Furthermore, although we used the JIF as the core explanatory variable, future studies could incorporate a broader set of metrics to better capture journals’ positions within a stratified publishing system.

We could not construct consistent longitudinal indicators of society ownership or journal geographic orientation for the full 1997–2022 period. Contemporary publisher information does not necessarily identify journal ownership, because commercial publishers frequently publish journals on behalf of scholarly societies, and ownership arrangements may change over time. Similarly, the country in which a publisher is headquartered does not necessarily represent a journal’s scholarly geographic orientation. We therefore avoided introducing potentially misleading contemporary proxies for these characteristics. Their stable components are absorbed by journal fixed effects in the within-journal analyses, but unobserved differences in ownership and geographic orientation may continue to confound the between-journal estimates.

Finally, JIF cannot be treated as an exogenous determinant of author-country concentration. The two variables may be jointly determined: journals’ author-country composition may influence

the geographic reach and volume of citations and hence subsequent JIF values, while journal policies, editorial networks, and other time-varying factors may affect both JIF and author composition. Journal and year fixed effects do not eliminate these sources of endogeneity. Consequently, our analyses establish a U-shaped empirical association but cannot determine its causal direction or identify the editorial, social or intellectual mechanisms underlying it. Future research using submission, peer-review, editorial-decision or credible quasi-experimental data is needed to evaluate these mechanisms and causal directions directly.

Author Contributions

Fan Zhang: writing – original draft. **Lili Wang:** writing – review and editing. **Xianwen Wang:** writing – review and editing. **Guangyao Zhang:** writing – review and editing, writing – original draft, visualization, software, methodology, formal analysis, conceptualization, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data will be made available on request.

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Appendix A

TABLE A1 | Annual values underlying Figure 1: International collaboration and team size, 1997–2022.

Publication year	Countries per first author	Unique countries per paper	Authors per paper
1997	1.01	1.07	1.62
1998	1.02	1.06	1.70
1999	1.01	1.06	1.67
2000	1.02	1.06	1.61
2001	1.01	1.06	1.70
2002	1.02	1.06	1.65
2003	1.01	1.07	1.67
2004	1.02	1.07	1.71
2005	1.02	1.08	1.74
2006	1.02	1.09	1.74
2007	1.02	1.11	1.80
2008	1.03	1.11	1.79
2009	1.02	1.11	1.81
2010	1.02	1.11	1.82
2011	1.02	1.13	1.88
2012	1.03	1.14	1.90
2013	1.03	1.14	1.95
2014	1.03	1.15	2.01
2015	1.03	1.15	1.98
2016	1.03	1.15	1.99
2017	1.02	1.16	2.00
2018	1.03	1.17	2.03
2019	1.03	1.18	2.09
2020	1.03	1.20	2.13
2021	1.03	1.21	2.22
2022	1.04	1.21	2.25

Note: Values are annual means. 'Countries per first author' measures the number of countries associated with the first author. 'Unique countries per paper' counts the distinct countries represented among a paper's authors after retaining each author's first listed country. The complete annual series constitute the supporting data for Figure 1.

TABLE A2 | Distribution of author-country concentration by JIF quartile.

JIF quartile	N	Mean	S.D.	Min	P25	Median	P75	Max
Q4	526	0.509	0.279	0.061	0.264	0.480	0.753	1.000
Q3	709	0.450	0.274	0.066	0.199	0.405	0.680	1.000
Q2	758	0.403	0.259	0.056	0.184	0.333	0.590	1.000
Q1	765	0.475	0.264	0.052	0.235	0.446	0.703	1.000

Note: The unit of analysis is the journal-year observation. Q1 denotes the highest JIF quartile and Q4 the lowest. P25 and P75 denote the 25th and 75th percentiles, respectively. Higher HHI values indicate greater author-country concentration. Outlier markers are suppressed in Figure 2 for visual clarity, whereas the minimum and maximum reported in this table refer to the full observed range.

TABLE A3 | Annual mean author-country concentration by JIF quartile, 1997–2022.

Year	Q1, mean HHI (N)	Q2, mean HHI (N)	Q3, mean HHI (N)	Q4, mean HHI (N)
1997	0.382 (21)	0.420 (21)	0.339 (19)	0.335 (10)
1998	0.670 (21)	0.508 (23)	0.582 (19)	0.429 (9)
1999	0.702 (23)	0.585 (22)	0.593 (22)	0.518 (8)
2000	0.659 (23)	0.529 (22)	0.501 (20)	0.635 (12)
2001	0.632 (23)	0.513 (23)	0.518 (20)	0.613 (13)
2002	0.655 (24)	0.510 (22)	0.606 (20)	0.420 (12)
2003	0.562 (23)	0.565 (23)	0.638 (20)	0.526 (13)
2004	0.654 (22)	0.519 (22)	0.572 (21)	0.493 (14)
2005	0.591 (23)	0.492 (24)	0.605 (21)	0.445 (15)
2006	0.594 (23)	0.417 (22)	0.541 (23)	0.567 (15)
2007	0.553 (24)	0.427 (23)	0.616 (21)	0.507 (14)
2008	0.610 (25)	0.346 (24)	0.543 (22)	0.649 (15)
2009	0.522 (28)	0.447 (27)	0.459 (25)	0.525 (16)
2010	0.462 (33)	0.380 (31)	0.468 (28)	0.441 (19)
2011	0.417 (34)	0.423 (32)	0.408 (29)	0.530 (24)
2012	0.426 (34)	0.405 (33)	0.400 (32)	0.542 (21)
2013	0.412 (34)	0.373 (34)	0.455 (32)	0.512 (22)
2014	0.434 (35)	0.390 (33)	0.391 (32)	0.420 (27)
2015	0.400 (35)	0.371 (35)	0.401 (32)	0.516 (25)
2016	0.380 (35)	0.384 (36)	0.343 (34)	0.556 (27)
2017	0.380 (36)	0.344 (36)	0.407 (37)	0.555 (29)
2018	0.372 (37)	0.328 (37)	0.379 (36)	0.522 (31)
2019	0.353 (37)	0.277 (38)	0.409 (37)	0.520 (33)
2020	0.388 (37)	0.313 (37)	0.359 (37)	0.550 (33)
2021	0.340 (37)	0.311 (38)	0.349 (37)	0.468 (34)
2022	0.376 (38)	0.308 (40)	0.314 (33)	0.435 (35)

Note: Q1 denotes the highest JIF quartile and Q4 the lowest. Entries report the annual mean HHI, with the number of contributing journals shown in parentheses. Higher HHI values indicate greater author-country concentration.

TABLE A4 | Numerical results underlying Figure 4: Within-journal association between annual JIF and author-country concentration.

Panel A. Quadratic fixed-effects estimates				
Variable	Coefficient		Standard error	Significance
Centred annual JIF	−0.048		0.023	$p < 0.05$
Squared centred annual JIF	0.007		0.002	$p < 0.01$
Journal fixed effects	Yes			
Year fixed effects	Yes			
Journal-year observations	2757			
Journals	150			
Panel B. Lind–Mehlum U-shape test				
Test component	Centred JIF	Original JIF	Slope/test statistic	p or 95% CI
Lower bound	−1.153	0.000	Slope = −0.064	$p = 0.010$
Turning point	3.673	4.825	—	95% CI: 3.076–6.575
Upper bound	12.047	13.200	Slope = 0.110	$p < 0.001$
Overall U-shape test	—	—	$t = 2.35$	$p = 0.010$

Note: Panel A reproduces the quadratic terms from the journal fixed-effects model reported in Column (3) of Table 4. The dependent variable is $\ln(\text{HHI})$. Panel B reports the numerical results underlying Figure 4. JIF is mean-centred in the estimation and figure using a centring constant of 1.153. Accordingly, the turning point of 3.673 on the centred scale corresponds to an annual JIF of 4.825 on the original scale. Standard errors are clustered by journal.

TABLE A5 | Robustness of the nonlinear association between JIF and author-country concentration to journal-year sample size.

Panel A. Restrictions and article-count weighting				
	(1)	(2)	(3)	(4)
	$N \geq 5$	$N \geq 10$	$N \geq 15$	Article-count weighted
Centred annual JIF	−0.022 (0.020)	−0.025 (0.019)	−0.021 (0.018)	−0.016 (0.019)
Squared centred annual JIF	0.004** (0.002)	0.004** (0.002)	0.004** (0.002)	0.004*** (0.001)
Gold OA ratio	0.322*** (0.092)	0.398*** (0.064)	0.464*** (0.156)	0.338*** (0.087)
ln(Total Papers)	−0.178*** (0.031)	−0.182*** (0.037)	−0.182*** (0.040)	−0.172*** (0.040)
Year FE	Yes	Yes	Yes	Yes
Journal FE	Yes	Yes	Yes	Yes
Constant	−1.004*** (0.115)	−1.184*** (0.180)	−0.901*** (0.230)	−0.523*** (0.150)
Observations	2698	2588	2406	2755
Within R^2	0.307	0.321	0.319	0.367
Panel B. Alternative construction of the dependent variable				
	(5)	(6)	(7)	
	Bias-corrected HHI, with N control	Bias-corrected HHI, without N control	Pooled 3-year HHI	
Centred annual JIF	−0.017* (0.010)	−0.018* (0.010)	−0.029 (0.027)	
Squared centred annual JIF	0.002* (0.001)	0.002** (0.001)	0.006*** (0.002)	
Gold OA ratio	0.136*** (0.020)	0.146*** (0.019)	0.221 (0.172)	
ln(Total Papers)	−0.023* (0.014)		−0.104*** (0.033)	
Year FE	Yes	Yes	Yes	
Journal FE	Yes	Yes	Yes	
Constant	0.255*** (0.047)	0.208*** (0.034)	−0.497*** (0.131)	
Observations	2737	2737	1010	
Within R^2	0.235	0.233	0.405	

Note: Columns (1–4) use the logarithm of annual HHI as the dependent variable. Column (4) weights journal-year observations by article count. Column (4) excludes two singleton observations automatically dropped when absorbing journal fixed effects. Columns (5) and (6) use the finite-sample-corrected HHI, $(N \times \text{HHI} - 1)/(N - 1)$, for $(N > 1)$. Column (7) uses the logarithm of HHI recalculated from article-level country shares pooled within non-overlapping 3-year periods. In this specification, Total Papers is the pooled article count, while JIF and the Gold OA ratio are averaged across the years in each period; the period-average JIF is centred before constructing its squared term. The final period covers 2021–2022. All models include journal and year or period fixed effects. Standard errors clustered by journal are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

TABLE A6 | Robustness to alternative measures of author-country concentration.

	(1)	(2)	(3)	(4)
	Entropy	Gini	Entropy	Gini
	Journal FE	Journal FE	Between-journal	Between-journal
Centred annual JIF	−0.558 (0.370)	−0.094** (0.037)		
Squared centred annual JIF	0.085** (0.036)	0.013*** (0.003)		
Gold OA ratio	5.812*** (1.504)	0.364* (0.202)		
ln(Total Papers)	−5.161*** (0.480)	−0.375*** (0.072)		
Mean JIF (centred)			−2.657* (1.540)	−0.152 (0.109)
Mean JIF sq.			2.461*** (0.613)	0.159*** (0.043)
Mean Gold OA ratio			1.195 (2.501)	0.069 (0.169)
Mean ln(Total Papers)			−6.448*** (1.505)	−0.585*** (0.148)
Year FE	Yes	Yes	No	No
Journal FE	Yes	Yes	No	No
Constant	86.230*** (1.743)	99.248*** (0.174)	96.212*** (4.996)	100.407*** (0.473)
Observations	2757	2757	150	150
Within R^2	0.401	0.353	—	—
R^2	—	—	0.217	0.243

Note: The dependent variables are entropy-based concentration in Columns (1) and (3) and the Gini coefficient in Columns (2) and (4). Both measures are expressed on a 0–100 scale, with higher values indicating greater author-country concentration. The entropy-based measure is calculated as $100 \times [1 - H/\ln(M)]$, where H is Shannon entropy and M is the fixed number of countries in the dataset. The Gini coefficient is calculated over the same fixed country universe, including zero shares for countries not represented in a journal-year. Columns (1) and (2) are journal fixed-effects models and report standard errors clustered by journal. Columns (3) and (4) are estimated from journal-level means and report heteroskedasticity-robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

TABLE A7 | Robustness to using the 5-year Journal Impact Factor.

	(1)	(2)
	Between-journal	Journal fixed effects
Mean JIF (centred)	−0.067 (0.058)	
Mean JIF sq.	0.037*** (0.012)	
Mean Gold OA ratio	0.225* (0.135)	
Mean ln(Total Papers)	−0.278** (0.115)	
Centred five-year JIF		−0.020 (0.022)
Squared centred five-year JIF		0.003** (0.001)
Gold OA ratio		0.414 (0.508)
ln(Total Papers)		−0.104*** (0.035)
Year_FE	No	Yes
Journal_FE	No	Yes
Constant	−0.231 (0.401)	−0.569*** (0.119)
Observations	147	1800
Within R^2	—	0.193
R^2	0.123	—

Note: The dependent variable is the logarithm of author-country HHI. Column (1) is estimated from journal-level means calculated over the journal-years for which the 5-year JIF is available. Column (2) is a journal fixed-effects model based on the corresponding journal-year observations. Five-year JIF data are unavailable before 2007 and are also missing for some subsequent journal-years; both models are therefore estimated on a common availability sample.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.